

Ingegneria Informatica e Robotica
Information and Estimation Theory

ESTIMATION THEORY

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References

- **Textbook**

  S. Kay, Fundamentals of Statistical Signal Processing Vol. I: Estimation Theory, Prentice Hall, 1993

- **These slides**

  Available in the Unistudium repository

- **Additional books**

  B. Levy, Principles of Signal Detection and Parameter Estimation, Springer, 2008

  P. Moulin, V. V. Veeravalli, Statistical Inference for Engineers and Data Scientists, Cambridge University Press, 2018

Motivation

- Given a set of data (observations) that depend on an unknown parameter, **an estimator tries to extract an unknown information from noisy data**, using signal processing techniques

- The applications are many; some of them are
  - Radar
  - Sonar
  - Speech
  - Image processing
  - Biomedicine
  - Communications
  - Control
  - Seismology



What is an estimator?

- Given a set of data $\{x[0], x[1], \ldots, x[N-1]\}$ that depend on an unknown parameter θ, an estimator is a function g of the data that produces an estimate of the unknown parameter:

$$\hat{\theta} = g(x[0], \ldots, x[N-1])$$

- The unknown parameter θ is scalar (or vector)

- There exist two approaches for estimation: **classical estimation** and **Bayesian estimation**



Classical estimation

- The true unknown parameter θ is assumed fixed and **deterministic**
- The data $\{x[0], x[1], \ldots, x[N-1]\}$ are modeled as random, with pdf $f_x(x[0],\ldots,x[N-1];\theta)$ that depends on the true unknown parameter θ
- The estimated parameter is a random variable:

$$\hat{\theta} = g(x[0],\ldots,x[N-1])$$

- The performance of the estimator is evaluated statistically (using the pdf/bias/variance of $\hat{\theta}$)

Bayesian estimation

- The true unknown parameter θ is assumed a **random variable** with known statistics (such as known pdf) that can be exploited for estimation

- The joint pdf of data and unknown parameter is $f_{x\theta}(x[0],...,x[N-1],\theta) = f_{x|\theta}(x[0],...,x[N-1]|\theta) f_{\theta}(\theta)$ where $f_{\theta}(\theta)$ is our prior knowledge about the unknown parameter θ

- Also in this case the estimated parameter $\hat{\theta}$ is random and the performance of the estimator is given by the pdf (or variance) of $\hat{\theta}$



PART 1

Classical Estimation



Classical estimation

- Minimum Variance Unbiased (MVU) Estimation
- Cramer-Rao Lower Bound (CRLB)
- Best Linear Unbiased Estimator (BLUE)
- Maximum Likelihood Estimator (MLE)
- Least Squares Estimator (LSE)



Estimation error

- The estimation error is the difference between the estimated parameter and the true parameter
- The estimation error is defined as

$$\varepsilon(\theta,\hat{\theta}) = \hat{\theta} - \theta$$

- It depends on the unknown true parameter θ
- It is a random variable, because depends on $\hat{\theta}$
- Scope: make $\varepsilon(\theta,\hat{\theta})$ as small as possible

Bias

- The **bias** is the mean value of the estimation error
- The bias is defined as

$$b(\theta) = E_{\hat{\theta}}\{\varepsilon(\theta,\hat{\theta})\} = E_{\hat{\theta}}\{\hat{\theta}\} - \theta$$

- An estimator is **unbiased** when its bias is zero

$$b(\theta) = 0 \;\Leftrightarrow\; E_{\hat{\theta}}\{\hat{\theta}\} = \theta$$

- Often, the estimator is required to be unbiased: in this case the estimated parameter is, on average, equal to the true parameter



Example: DC level

- The data model is $x[n] = \theta + w[n], \quad n = 0, \ldots, N-1,$

  $w[n]$ is a zero-mean additive white Gaussian noise (AWGN), uncorrelated

- Possible estimators are: single sample $\hat{\theta}^{(1)} = x[0]$,

  sample mean $\hat{\theta}^{(2)} = \frac{1}{N} \sum_{n=0}^{N-1} x[n]$

- Which one is better? We have to compare the performance of the two estimators ➔ comparison in terms of pdf/mean/variance

Example: DC level (bias)

- $x[n] = \theta + w[n], \quad n = 0, \ldots, N-1, \qquad w[n]$ AWGN

- Let us consider $\hat{\theta}^{(1)} = x[0]$ and $\hat{\theta}^{(2)} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]$

$$b_1(\theta) = E\{x[0]\} - \theta = (\theta + E\{w[0]\}) - \theta = \theta - \theta = 0$$

$$b_2(\theta) = E\left\{\frac{1}{N}\sum_{n=0}^{N-1} x[n]\right\} - \theta = \frac{1}{N}\sum_{n=0}^{N-1} E\{x[n]\} - \theta =$$

$$= \frac{1}{N}\sum_{n=0}^{N-1} (\theta + E\{w[n]\}) - \theta = \frac{1}{N} N\theta - \theta = \theta - \theta = 0$$

- Both estimators are unbiased



Reminder

- Definition of mean value:

$$E_{\hat{\theta}}\{\hat{\theta}\} = \int \hat{\theta} f_{\hat{\theta}}(\hat{\theta}) d\hat{\theta}$$

- Definition of variance:

$$\mathrm{Var}\{\hat{\theta}\} = \int \left(\hat{\theta} - E_{\hat{\theta}}\{\hat{\theta}\}\right)^2 f_{\hat{\theta}}(\hat{\theta}) d\hat{\theta} =$$

$$= \int \hat{\theta}^2 f_{\hat{\theta}}(\hat{\theta}) d\hat{\theta} - \left(E_{\hat{\theta}}\{\hat{\theta}\}\right)^2$$

Variance

- The **variance** of an estimator tells how much the estimated parameter is close to its mean
- Often, the estimator is required to have low variance: in this case the estimated parameter is, on average, close to the mean estimated value
- When the estimator is unbiased, a low variance implies that the estimated parameter is close to the true parameter
- When the estimator is biased, a low variance implies either overestimation or underestimation



Example: DC level (variance)

- $x[n] = \theta + w[n], \quad n = 0, \ldots, N-1, \qquad w[n]$ AWGN

- Let us consider $\hat{\theta}^{(1)} = x[0]$ and $\hat{\theta}^{(2)} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]$

$$\mathrm{Var}\{\hat{\theta}^{(1)}\} = \mathrm{Var}\{x[0]\} = \mathrm{Var}\{w[0]\} = \sigma_w^2$$

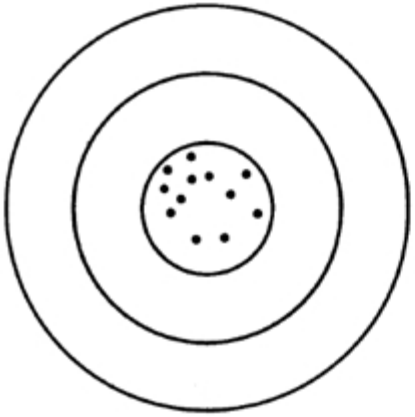
$$\mathrm{Var}\{\hat{\theta}^{(2)}\} = \mathrm{Var}\left\{\frac{1}{N}\sum_{n=0}^{N-1} x[n]\right\} = \frac{1}{N^2}\mathrm{Var}\left\{\sum_{n=0}^{N-1} x[n]\right\} =$$

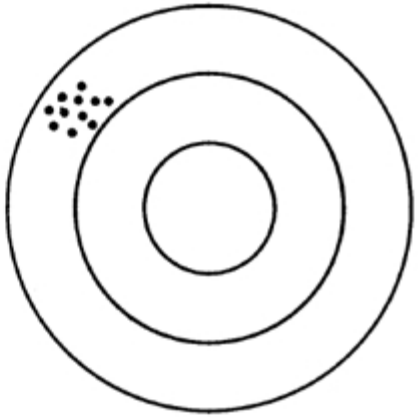
$$= \frac{1}{N^2}\sum_{n=0}^{N-1}\mathrm{Var}\{x[n]\} = \frac{1}{N^2} N\sigma_w^2 = \frac{\sigma_w^2}{N}$$

- Sample mean is better (variance is N times lower)

Bias and variance

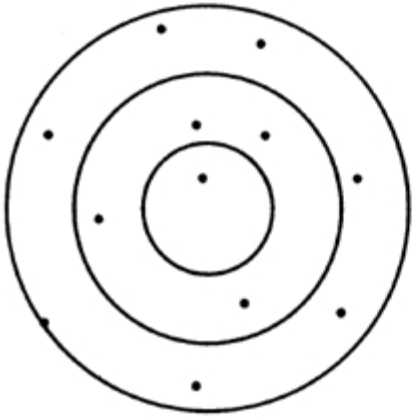
low bias
and
low variance

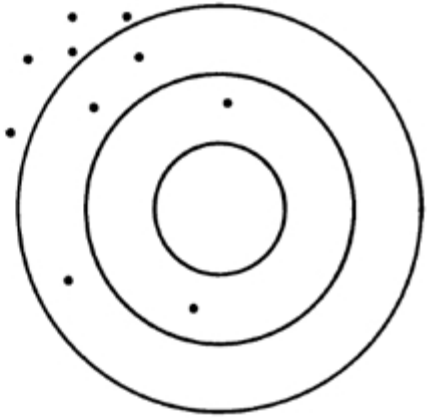




large bias
and
low variance

low bias
and
large variance





large bias
and
large variance

How to find a good estimator?

- To find a good estimator ➔ many criteria exist ➔ we would like both low bias and low variance ➔ we may minimize the mean square error (MSE)

- The MSE is the power of the estimation error:

$$\begin{aligned}\mathrm{MSE}\{\hat{\theta}\} &= E_{\hat{\theta}}\{\varepsilon^2(\theta,\hat{\theta})\} = \\ &= \int (\hat{\theta}-\theta)^2 f_{\hat{\theta}}(\hat{\theta}) d\hat{\theta} = \\ &= \mathrm{Var}\{\hat{\theta}\} + b^2(\theta)\end{aligned}$$

- Low MSE implies both low bias and low variance



Proof of $\mathrm{MSE}\{\hat{\theta}\} = \mathrm{Var}\{\hat{\theta}\} + b^2(\theta)$

- Let $\mu_{\hat{\theta}} = E_{\hat{\theta}}\{\hat{\theta}\}$, then we have

$$\mathrm{MSE}\{\hat{\theta}\} = E_{\hat{\theta}}\{(\hat{\theta}-\theta)^2\} =$$

$$= E_{\hat{\theta}}\{(\hat{\theta}-\mu_{\hat{\theta}}+\mu_{\hat{\theta}}-\theta)^2\} =$$

$$= E_{\hat{\theta}}\{(\hat{\theta}-\mu_{\hat{\theta}})^2 + 2(\hat{\theta}-\mu_{\hat{\theta}})(\mu_{\hat{\theta}}-\theta) + (\mu_{\hat{\theta}}-\theta)^2\} =$$

$$= E_{\hat{\theta}}\{(\hat{\theta}-\mu_{\hat{\theta}})^2\} + E_{\hat{\theta}}\{2(\hat{\theta}-\mu_{\hat{\theta}})(\mu_{\hat{\theta}}-\theta)\} + E_{\hat{\theta}}\{(\mu_{\hat{\theta}}-\theta)^2\} =$$

$$= \mathrm{Var}\{\hat{\theta}\} + 2E_{\hat{\theta}}\{\hat{\theta}-\mu_{\hat{\theta}}\}(\mu_{\hat{\theta}}-\theta) + (\mu_{\hat{\theta}}-\theta)^2 =$$

$$= \mathrm{Var}\{\hat{\theta}\} + 2(E_{\hat{\theta}}\{\hat{\theta}\}-\mu_{\hat{\theta}})b(\theta) + b^2(\theta) =$$

$$= \mathrm{Var}\{\hat{\theta}\} + 2\cdot 0\cdot b(\theta) + b^2(\theta) =$$

$$= \mathrm{Var}\{\hat{\theta}\} + b^2(\theta)$$



1.1

Minimum Variance Unbiased (MVU) Estimation

Minimum Variance Unbiased estimation

- To reduce both bias and variance ➔ we may try to minimize the MSE

$$\mathrm{MSE}\{\hat{\theta}\} = \mathrm{Var}\{\hat{\theta}\} + b^2(\theta)$$

- In many cases, the bias $b(\theta)$ depends on the unknown parameter θ ➔ in classical estimation, <u>the minimization of the MSE is often impossible</u> because requires the unknown parameter θ

- Solution: force bias to zero ➔ minimize variance ➔ **minimum variance unbiased** (MVU) estimation



Finding the MVU estimator

- Finding the MVU estimator (unbiased estimator with minimum variance) is not easy ➔ the MVU estimator may not exist ➔ and when the MVU exists, there is no general strategy to find it

- To find an MVU estimator ➔ two strategies are:

  1. Find the Cramer-Rao Lower Bound (CRLB) and check if some estimator satisfies the bound

  2. Restrict the search within the subset of linear estimators ➔ Best Linear Unbiased Estimator (BLUE)

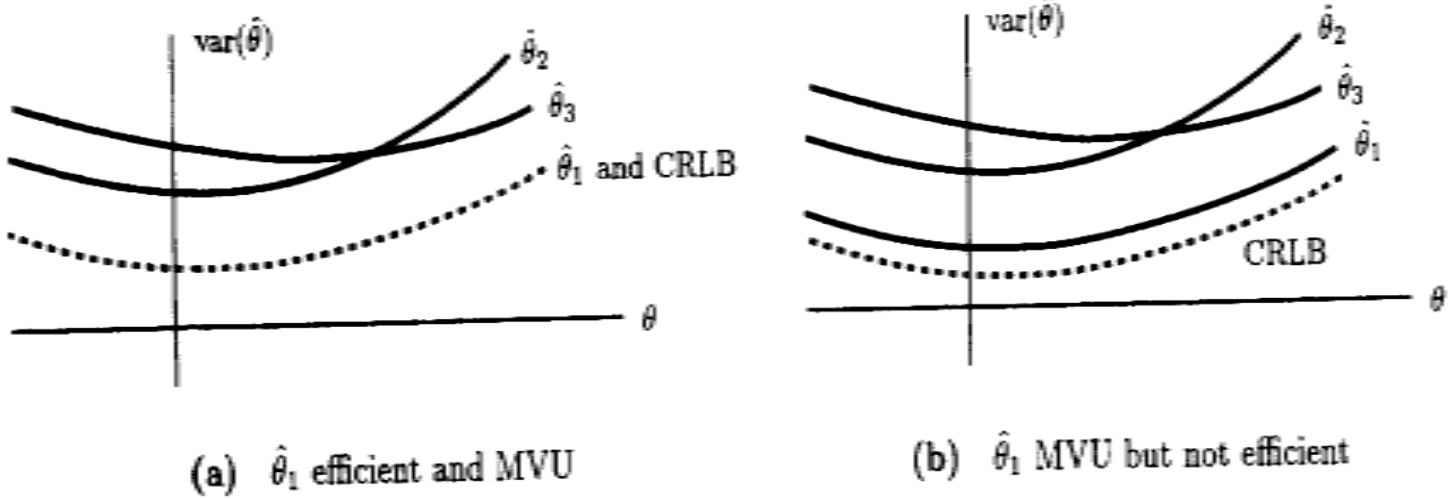


1.2

Cramer-Rao Lower Bound (CRLB)

Cramer-Rao lower bound (CRLB)

- The CRLB is a lower bound on the variance of any unbiased estimator ➔ bound on the performance



(a) $\hat{\theta}_1$ efficient and MVU

(b) $\hat{\theta}_1$ MVU but not efficient

- In some cases, it is possible to find an estimator that satisfies the CRLB ➔ in these cases, the estimator is also the MVU estimator ➔ an unbiased estimator that reaches the CRLB is said **efficient**

CRLB theorem

- If the pdf $f_x(x;\theta) = f_x(x[0],\ldots,x[N-1];\theta)$ satisfies the regularity condition

$$E_x\left\{\frac{\partial \ln f_x(x;\theta)}{\partial\theta}\right\} = 0, \quad \forall\theta$$

then the variance of any unbiased estimator is bounded by

$$\mathrm{Var}\{\hat{\theta}\} \geq \frac{1}{I(\theta)}$$

where $I(\theta)$ is the **Fisher information**, given by

$$I(\theta) = -E_x\left\{\frac{\partial^2 \ln f_x(x;\theta)}{\partial\theta^2}\right\} = E_x\left\{\left(\frac{\partial \ln f_x(x;\theta)}{\partial\theta}\right)^2\right\}$$



CRLB theorem

- An unbiased estimator that achieves the CRLB can be obtained if and only if

$$\frac{\partial \ln f_x(x;\theta)}{\partial \theta} = I(\theta)[g(x) - \theta]$$

When this is possible, the estimator $\hat{\theta}_{\text{MVU}} = g(x)$ is the MVU estimator and its variance is

$$\text{Var}\{\hat{\theta}_{\text{MVU}}\} = \frac{1}{I(\theta)}$$

- The CRLB theorem gives a procedure for finding the MVU estimator



Example: CRLB for DC level

- $x[n] = \theta + w[n], \quad n = 0, \ldots, N-1, \qquad w[n]$ AWGN
- 1st step: pdf of the data

$$f_x(x;\theta) = f_x(x[0], \ldots, x[N-1];\theta) =$$

$$= \frac{1}{(2\pi\sigma_w^2)^{N/2}} \exp\left[-\frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x[n]-\theta)^2 \right]$$

- 2nd step: logarithm of the pdf

$$\ln f_x(x;\theta) = -\frac{N}{2} \ln(2\pi\sigma_w^2) - \frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x[n]-\theta)^2$$



Example: CRLB for DC level

- 3rd step: derivative with respect to θ

$$\frac{\partial \ln f_x(x;\theta)}{\partial \theta} = \frac{1}{\sigma_w^2}\sum_{n=0}^{N-1}(x[n]-\theta)$$

- 4th step: check the regularity condition

$$E_x\left\{\frac{\partial \ln f_x(x;\theta)}{\partial \theta}\right\} = \frac{1}{\sigma_w^2}\sum_{n=0}^{N-1}(E_x\{x[n]\}-\theta) = \frac{1}{\sigma_w^2}\sum_{n=0}^{N-1}(\theta-\theta) = 0$$

- 5th step: second derivative

$$\frac{\partial^2 \ln f_x(x;\theta)}{\partial \theta^2} = -\frac{N}{\sigma_w^2}$$



Example: CRLB for DC level

- 6th step: Fisher information

$$I(\theta) = -E_x\left\{\frac{\partial^2 \ln f_x(x;\theta)}{\partial\theta^2}\right\} = \frac{N}{\sigma_w^2}$$

- 7th step: find $g(x)$, using the result of 3rd step

$$\frac{\partial \ln f_x(x;\theta)}{\partial\theta} = \frac{1}{\sigma_w^2}\left[\left(\sum_{n=0}^{N-1} x[n]\right) - N\theta\right] = \frac{N}{\sigma_w^2}\left[\left(\frac{1}{N}\sum_{n=0}^{N-1} x[n]\right) - \theta\right]$$

- Hence **the MVU estimator is the sample mean**

$$\hat{\theta}_{\mathrm{MVU}} = g(x) = \frac{1}{N}\sum_{n=0}^{N-1} x[n] \qquad \mathrm{Var}\{\hat{\theta}_{\mathrm{MVU}}\} = \frac{1}{I(\theta)} = \frac{\sigma_w^2}{N}$$

Transformation of parameters

- If we know the CRLB for the estimation of θ ➔ we can obtain the CRLB for $\alpha = h(\theta)$ as

$$\mathrm{Var}\{\hat{\alpha}\} \geq \frac{\left(\frac{\partial h}{\partial \theta}\right)^2}{I(\theta)}$$

- If $\hat{\theta} = g(x)$ is an efficient estimator of θ, the estimator $\hat{\alpha} = h(\hat{\theta}) = h(g(x))$ is not necessarily efficient (could be biased) ➔ however usually it is asymptotically efficient (when $N \to \infty$, the estimator becomes unbiased and efficient)

Transformation of parameters

- If we know the CRLB for the estimation of θ ➔ we can obtain the CRLB for $\alpha = k\theta + c$ as

$$\mathrm{Var}\{\hat{\alpha}\} \geq \frac{k^2}{I(\theta)}$$

- If $\hat{\theta} = g(x)$ is an efficient estimator of θ, also the estimator $\hat{\alpha} = k\hat{\theta} + c = kg(x) + c$ is efficient ➔ **efficiency is preserved by affine or linear transformations of the parameter**

- Nonlinear transformations do not preserve efficiency, they only preserve asymptotical efficiency



Example: gain from DC level

- $x[n] = \theta + w[n], \quad n = 0, \ldots, N-1, \qquad w[n]$ AWGN
- To estimate the gain $\alpha = \theta^2$ ➔ possible estimator:

$$\hat{\alpha} = \left(\hat{\theta}\right)^2 = \left(\frac{1}{N}\sum_{n=0}^{N-1} x[n]\right)^2$$

- Bias $b(\alpha) = \dfrac{\sigma_w^2}{N}$ ➔ bias vanishes for large N ➔ the estimator is asymptotically unbiased
- $\alpha = h(\theta) = \theta^2$ ➔ CRLB of α is obtained from the CRLB of θ



Example: gain from DC level

- $\frac{\partial h}{\partial \theta} = 2\theta$ and $I(\theta) = \frac{N}{\sigma_w^2}$ give the CRLB

$$\mathrm{Var}\{\hat{\alpha}\} \geq \frac{(2\theta)^2}{I(\theta)} = \frac{4\alpha\sigma_w^2}{N}$$

- The variance of the estimator is actually

$$\mathrm{Var}\{\hat{\alpha}\} = \frac{4\alpha\sigma_w^2}{N} + \frac{2\sigma_w^4}{N^2}$$

➔ for large N, the second term goes to zero faster than the first term ➔ variance tends to the CRLB ➔ estimator is asymptotically efficient

CRLB theorem – vector case

- If the pdf $f_x(x;\boldsymbol{\theta}) = f_x(x[0],\ldots,x[N-1];\theta_1,\ldots,\theta_P)$ satisfies the regularity conditions

$$E_x\left\{\frac{\partial \ln f_x(x;\boldsymbol{\theta})}{\partial \theta_i}\right\} = 0, \quad \forall \theta_i$$

then the covariance matrix $\mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}}$ of any unbiased estimator $\hat{\boldsymbol{\theta}}$ satisfies the positive semidefinite inequality

$$\mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}} - \mathbf{I}^{-1}(\boldsymbol{\theta}) \geq 0$$

where $\mathbf{I}(\boldsymbol{\theta})$ is the Fisher information matrix (FIM), given by

$$[\mathbf{I}(\boldsymbol{\theta})]_{i,j} = -E_x\left\{\frac{\partial^2 \ln f_x(x;\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j}\right\}$$



CRLB theorem – vector case

- An unbiased estimator that achieves the CRLB can be obtained if and only if

$$\frac{\partial \ln f_x(x;\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{I}(\boldsymbol{\theta})[\mathbf{g}(x) - \boldsymbol{\theta}]$$

- When this is possible, the estimator $\hat{\boldsymbol{\theta}} = \mathbf{g}(x)$ is the MVU estimator and its covariance is

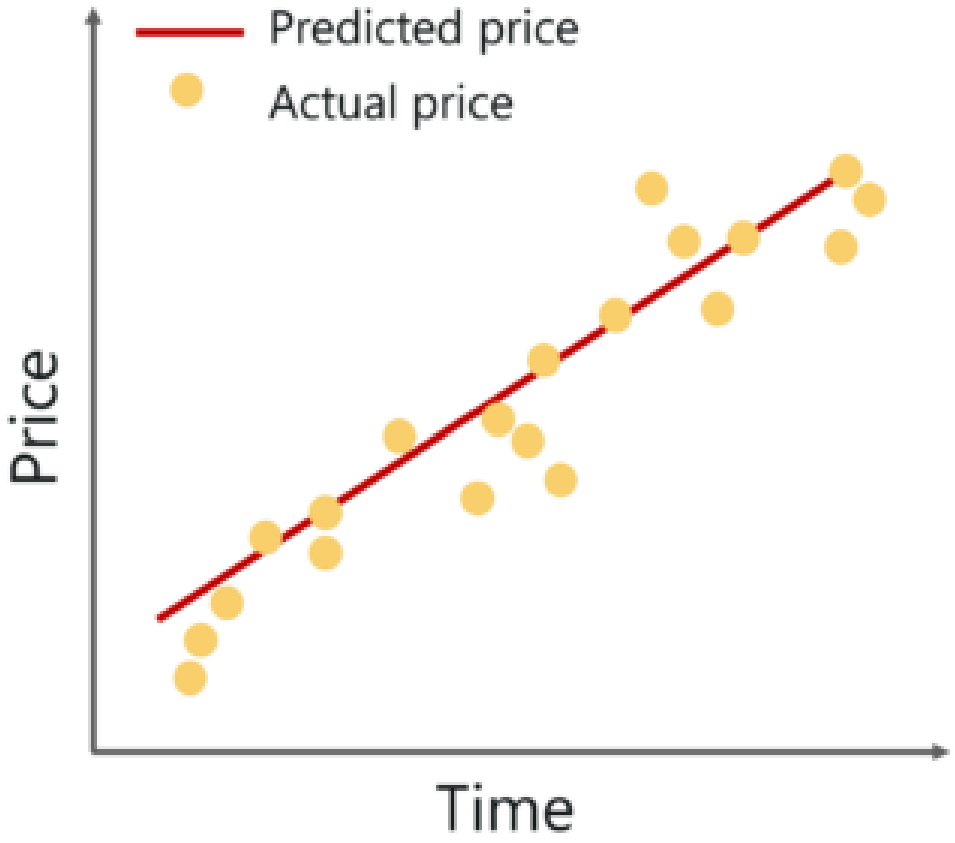
$$\mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}} = \mathbf{I}^{-1}(\boldsymbol{\theta})$$

- In this case, the variance of the *i*th estimated parameter is

$$\mathrm{Var}\{\hat{\theta}_i\} = [\mathbf{I}^{-1}(\boldsymbol{\theta})]_{i,i}$$



Example: line fitting



Example: line fitting

- $x[n] = \theta_1 + \theta_2 n + w[n], \quad n = 0, \ldots, N-1$ $\quad\quad$ $w[n]$ AWGN
- $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$, pdf of the data and its logarithm are

$$f_x(x;\boldsymbol{\theta}) = \frac{1}{(2\pi\sigma_w^2)^{N/2}} \exp\left[-\frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x[n] - \theta_1 - \theta_2 n)^2 \right]$$

$$\ln f_x(x;\boldsymbol{\theta}) = -\frac{N}{2} \ln(2\pi\sigma_w^2) - \frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x[n] - \theta_1 - \theta_2 n)^2$$

- The FIM is obtained as

$$[\mathbf{I}(\boldsymbol{\theta})]_{i,j} = -E_x \left\{ \frac{\partial^2 \ln f_x(x;\boldsymbol{\theta})}{\partial\theta_i \partial\theta_j} \right\}$$

$$\mathbf{I}(\boldsymbol{\theta}) = \frac{1}{\sigma_w^2} \begin{bmatrix} N & \sum_{n=0}^{N-1} n \\ \sum_{n=0}^{N-1} n & \sum_{n=0}^{N-1} n^2 \end{bmatrix}$$

Example: line fitting

$$\mathbf{I}(\boldsymbol{\theta}) = \frac{1}{\sigma_w^2}\begin{bmatrix} N & \sum_{n=0}^{N-1} n \\ \sum_{n=0}^{N-1} n & \sum_{n=0}^{N-1} n^2 \end{bmatrix} = \frac{1}{\sigma_w^2}\begin{bmatrix} N & \frac{N(N-1)}{2} \\ \frac{N(N-1)}{2} & \frac{N(N-1)(2N-1)}{6} \end{bmatrix}$$

- FIM inverse is $\mathbf{I}^{-1}(\boldsymbol{\theta}) = \sigma_w^2 \begin{bmatrix} \frac{2(2N-1)}{N(N+1)} & -\frac{6}{N(N+1)} \\ -\frac{6}{N(N+1)} & \frac{12}{N(N^2-1)} \end{bmatrix}$

- Hence $\mathrm{Var}\{\hat{\theta}_1\} \geq \frac{2(2N-1)\sigma_w^2}{N(N+1)}$, $\mathrm{Var}\{\hat{\theta}_2\} \geq \frac{12\sigma_w^2}{N(N^2-1)}$

  ➔ for large N ➔ θ_2 is easier to estimate than θ_1



Example: line fitting

- We have $$\frac{\partial \ln f_x(x;\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \begin{bmatrix} \frac{1}{\sigma_w^2}\sum_{n=0}^{N-1}(x[n]-\theta_1-\theta_2 n) \\ \frac{1}{\sigma_w^2}\sum_{n=0}^{N-1}(x[n]-\theta_1-\theta_2 n)n \end{bmatrix}$$
- We also have $$\frac{\partial \ln f_x(x;\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{I}(\boldsymbol{\theta})[\mathbf{g}(x)-\boldsymbol{\theta}]$$, where

$$\mathbf{g}(x) = \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix} = \begin{bmatrix} \frac{2(2N-1)}{N(N+1)}\sum_{n=0}^{N-1}x[n] - \frac{6}{N(N+1)}\sum_{n=0}^{N-1}nx[n] \\ -\frac{6}{N(N+1)}\sum_{n=0}^{N-1}x[n] + \frac{12}{N(N^2-1)}\sum_{n=0}^{N-1}nx[n] \end{bmatrix}$$

- The CRLB holds with equality ➔ both estimators are MVU (it can be shown that the regularity conditions are satisfied, so they are unbiased)



1.3

Best Linear Unbiased Estimator (BLUE)

Best Linear Unbiased Estimator

- In some cases, the MVU estimator does not exist or may be difficult to find ➔ in other cases, the pdf of the data may be unknown ➔ we can restrict the estimator to be **linear in the data**

- With the linear constraint, we can search for the unbiased estimator with minimum variance ➔ best linear unbiased estimator (BLUE)

- The BLUE only requires the mean and covariance of the data ➔ the whole pdf of the data may be unknown

BLUE – unbiased

- $\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] = \mathbf{a}^T \mathbf{x}$, $\mathbf{x} = [x[0], x[1], ..., x[N-1]]^T$ data

  $\mathbf{a} = [a_0, a_1, ..., a_{N-1}]^T$ parameters

- Unbiased: $E_{\hat{\theta}}\{\hat{\theta}\} = \theta \Leftrightarrow E_x\{\mathbf{a}^T \mathbf{x}\} = \mathbf{a}^T E_x\{\mathbf{x}\} = \theta$

- This is possible only when $\mathbf{x} = \theta \mathbf{s} + \mathbf{w}$ ➔ **the data model must be linear** with respect to θ:

$$\mathbf{a}^T E_w\{\theta \mathbf{s} + \mathbf{w}\} = \theta \Leftrightarrow \mathbf{a}^T \mathbf{s} \theta = \theta \Leftrightarrow \mathbf{a}^T \mathbf{s} = 1$$

- $\mathbf{a}^T \mathbf{s} = 1$ is a constraint on the design vector $\mathbf{a}$

BLUE – minimum variance

- $\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] = \mathbf{a}^T \mathbf{x}, \qquad \mathbf{x} = \theta \mathbf{s} + \mathbf{w}, \qquad \mathbf{C}_{\mathbf{xx}} = \mathbf{C}_{\mathbf{ww}}$
- $\mathrm{Var}\{\hat{\theta}\} = E_{\hat{\theta}}\{(\hat{\theta} - E_{\hat{\theta}}\{\hat{\theta}\})^2\} = E_x\{[\mathbf{a}^T(\mathbf{x} - E_x\{\mathbf{x}\})]^2\} =$

  $= E_x\{\mathbf{a}^T(\mathbf{x} - E_x\{\mathbf{x}\})(\mathbf{x} - E_x\{\mathbf{x}\})^T\mathbf{a}\} = \mathbf{a}^T\mathbf{C}_{\mathbf{xx}}\mathbf{a}$
- BLUE ➔ minimize $\mathbf{a}^T\mathbf{C}_{\mathbf{xx}}\mathbf{a}$ subject to $\mathbf{a}^T\mathbf{s} = 1$
- Lagrange method: $J(\lambda) = \mathbf{a}^T\mathbf{C}_{\mathbf{xx}}\mathbf{a} + \lambda(\mathbf{a}^T\mathbf{s} - 1)$
- Solution: $\frac{\partial J(\lambda)}{\partial \mathbf{a}} = \mathbf{0} \Leftrightarrow 2\mathbf{C}_{\mathbf{xx}}\mathbf{a} + \lambda\mathbf{s} = \mathbf{0} \Leftrightarrow \mathbf{a} = -\frac{\lambda}{2}\mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{s}$

BLUE – solution

- Lagrange multiplier obtained from the constraint

$$\mathbf{a}^T\mathbf{s}=1 \;\Leftrightarrow\; -\frac{\lambda}{2}\mathbf{s}^T\mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{s}=1 \;\Leftrightarrow\; \lambda=-\frac{2}{\mathbf{s}^T\mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{s}}$$

- By inserting the multiplier in the solution ➔

$$\mathbf{a}=\frac{1}{\mathbf{s}^T\mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{s}}\mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{s}$$

- Hence the BLUE and its variance are

$$\hat{\theta}_{\mathrm{BLUE}}=\mathbf{a}^T\mathbf{x}=\frac{\mathbf{s}^T\mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{x}}{\mathbf{s}^T\mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{s}} \qquad \mathrm{Var}\{\hat{\theta}_{\mathrm{BLUE}}\}=\mathbf{a}^T\mathbf{C}_{\mathbf{xx}}\mathbf{a}=\frac{1}{\mathbf{s}^T\mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{s}}$$

Example: DC with colored noise

- $x[n] = \theta + w[n], \quad n = 0, \ldots, N-1, \quad \{w[n]\}$ independent with variance $\sigma_w^2[n]$ that changes with n

- $\mathbf{s} = [1, \ldots, 1]^T = \mathbf{1}_N$ ➔ constraint is $\mathbf{a}^T \mathbf{s} = 1 \Leftrightarrow \mathbf{a}^T \mathbf{1}_N = 1$

$$\sum_{n=0}^{N-1} a_n = 1$$

- Data covariance:

$$\mathbf{C}_{\mathbf{xx}} = \mathbf{C}_{\mathbf{ww}} = \begin{bmatrix} \sigma_w^2[0] & & 0 \\ & \ddots & \\ 0 & & \sigma_w^2[N-1] \end{bmatrix}$$

- Hence the BLUE is given by $\hat{\theta}_{\text{BLUE}} = \dfrac{\mathbf{1}_N^T \mathbf{C}_{\mathbf{xx}}^{-1} \mathbf{x}}{\mathbf{1}_N^T \mathbf{C}_{\mathbf{xx}}^{-1} \mathbf{1}_N} = \dfrac{\sum_{n=0}^{N-1} \dfrac{x[n]}{\sigma_w^2[n]}}{\sum_{n=0}^{N-1} \dfrac{1}{\sigma_w^2[n]}}$

BLUE – vector case

- $\hat{\theta}_i = \sum_{n=0}^{N-1} a_{i,n} x[n]$,

$\mathbf{x} = [x[0], x[1], ..., x[N-1]]^T$ data

$\hat{\boldsymbol{\theta}} = [\hat{\theta}_0, \hat{\theta}_1, ..., \hat{\theta}_{P-1}]^T$ estimators

$\hat{\boldsymbol{\theta}} = \mathbf{A}\mathbf{x}$

$$\mathbf{A} = \begin{bmatrix} a_{0,0} & \cdots & a_{0,N-1} \\ \vdots & & \vdots \\ a_{P-1,0} & \cdots & a_{P-1,N-1} \end{bmatrix}$$ parameters

- Unbiased when $E_{\hat{\theta}}\{\hat{\boldsymbol{\theta}}\} = \boldsymbol{\theta} \Leftrightarrow \mathbf{A}E_x\{\mathbf{x}\} = \boldsymbol{\theta}$

- This is possible only when $\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$ ➔ the data model must be linear ➔ the constraint is

$$\mathbf{A}E_w\{\mathbf{H}\boldsymbol{\theta} + \mathbf{w}\} = \boldsymbol{\theta} \Leftrightarrow \mathbf{A}\mathbf{H}\boldsymbol{\theta} = \boldsymbol{\theta} \Leftrightarrow \mathbf{A}\mathbf{H} = \mathbf{I}_P$$

BLUE – vector case

- $\hat{\boldsymbol{\theta}} = \mathbf{A}\mathbf{x}, \qquad \mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}, \qquad \mathbf{C}_{\mathbf{xx}} = \mathbf{C}_{\mathbf{ww}}$
- $\mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}} = \mathrm{Cov}\{\hat{\boldsymbol{\theta}}\} = \mathbf{A}\mathbf{C}_{\mathbf{xx}}\mathbf{A}^T$
- BLUE ➔ minimize $\mathbf{A}\mathbf{C}_{\mathbf{xx}}\mathbf{A}^T$ subject to $\mathbf{A}\mathbf{H} = \mathbf{I}_P$
- The solution is $\mathbf{A} = (\mathbf{H}^T\mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}_{\mathbf{xx}}^{-1}$
- The BLUE is $\hat{\boldsymbol{\theta}} = \mathbf{A}\mathbf{x} = (\mathbf{H}^T\mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{x}$
- The covariance is $\mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}} = (\mathbf{H}^T\mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{H})^{-1}$



Properties of the BLUE

- The BLUE is the best estimator (in the MVU sense) within the class of linear estimators

- When the noise is Gaussian, the BLUE coincides with the general MVU estimator ➔ the CRLB is satisfied because the FIM is $\mathbf{I}(\boldsymbol{\theta}) = \mathbf{H}^T \mathbf{C}_{\mathbf{xx}}^{-1} \mathbf{H} = \mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}}^{-1}$

- When the noise is white $\mathbf{C}_{\mathbf{ww}} = \sigma_w^2 \mathbf{I}_N$ ➔ the BLUE becomes $\hat{\boldsymbol{\theta}} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x}$ ➔ this coincides with the least squares solution, which does not use any probabilistic assumption ➔ in this case, the covariance of the estimator is $\mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}} = \sigma_w^2 (\mathbf{H}^T \mathbf{H})^{-1}$



1.4

Maximum Likelihood Estimator (MLE)

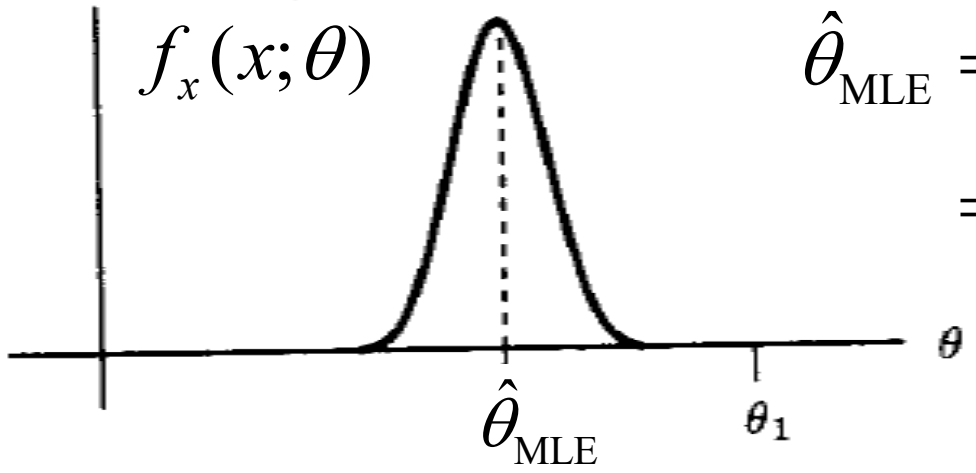
Maximum Likelihood Estimator

- MVU estimator may not exist or may be difficult to find, while BLUE is restricted to linear models ➔ try the **Maximum Likelihood Estimator** (MLE)
- When the pdf of the data is known, **the MLE can always be found**
- **MLE is optimal for large data size ➔** if an efficient estimator exists, the MLE will produce it
- Finding the MLE may be computationally complex and may require numerical methods

MLE principle

- MLE ➔ choose $\hat{\theta}$ that maximizes the probability of the observed data x

- The MLE maximizes the likelihood function ➔ the likelihood function is $f_x(x;\theta)$, the pdf of the data regarded as a function of θ ➔



$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} f_x(x;\theta) =$$

$$= \arg\max_{\theta} \ln f_x(x;\theta)$$



Finding the MLE

- 1st step: find the likelihood function $f_x(x;\theta)$
- 2st step: compute the log-likelihood function (LLF) $\ln f_x(x;\theta)$
- 3rd step: compute the derivative of the LLF with respect to θ and set it to zero

$$\frac{\partial}{\partial\theta}\ln f_x(x;\theta)=0$$

- 4th step: solve for θ the above equation, using an analytical/numerical method

Example: DC level equal to the noise variance

- $x[n] = \theta + \underbrace{w[n]}_{\text{AWGN}}, \quad n = 0, \ldots, N-1, \quad \sigma_w^2 = \theta \quad \text{with} \quad \theta > 0$
- Likelihood function, LLF, and derivative:

$$f_x(x;\theta) = \frac{1}{(2\pi\theta)^{N/2}} \exp\left[-\frac{1}{2\theta} \sum_{n=0}^{N-1} (x[n]-\theta)^2 \right]$$

$$\ln f_x(x;\theta) = -\frac{N}{2} \ln(2\pi\theta) - \frac{1}{2\theta} \sum_{n=0}^{N-1} (x[n]-\theta)^2$$

$$\frac{\partial}{\partial\theta} \ln f_x(x;\theta) = -\frac{N}{2} \frac{2\pi}{2\pi\theta} - \frac{1}{4\theta^2} \left[-2\theta \cdot 2 \sum_{n=0}^{N-1} (x[n]-\theta) - 2 \sum_{n=0}^{N-1} (x[n]-\theta)^2 \right] =$$

$$= \frac{1}{2\theta^2} \left[-N\theta^2 - N\theta + \sum_{n=0}^{N-1} x^2[n] \right]$$

Example: DC level equal to the noise variance

- $\frac{\partial}{\partial\theta}\ln f_x(x;\theta)=0 \quad \rightarrow \quad -N\theta^2-N\theta+\sum_{n=0}^{N-1}x^2[n]=0$

$$\theta^2+\theta-\frac{1}{N}\sum_{n=0}^{N-1}x^2[n]=0$$

- Solution:

$$\hat{\theta}_{\mathrm{MLE}}=-\frac{1}{2}+\sqrt{\frac{1}{4}+\frac{1}{N}\sum_{n=0}^{N-1}x^2[n]}$$

- Positive, asymptotically unbiased, asymptotically efficient ➔ CRLB is

$$\mathrm{Var}\{\hat{\theta}\}\geq\frac{2\theta^2}{(2\theta+1)N}$$



Properties of the MLE

- The MLE is asymptotically unbiased
- The MLE asymptotically achieves the CRLB ➔ **the MLE is asymptotically efficient** ➔ if the MVU estimator exists, the MLE procedure is a way to find the MVU estimator (for large data records)
- The pdf of the MLE is asymptotically Gaussian ➔ hence, for large N, $f_{\hat{\theta}_{\mathrm{MLE}}}(\hat{\theta}_{\mathrm{MLE}})$ tends to a Gaussian random variable with mean θ and variance $I^{-1}(\theta)$

Transformation of parameters

- **Invariance property**: if we know the MLE of θ ➔ we can obtain the MLE for $\alpha = h(\theta)$ as

$$\hat{\alpha}_{\mathrm{MLE}} = h(\hat{\theta}_{\mathrm{MLE}})$$

- This property is useful because the computation of the MLE of θ is usually simpler than the direct computation of the MLE of α based on the data pdf as a function of α

- Asymptotic efficiency is preserved

MLE – vector case

- MLE for a vector of parameters $\boldsymbol{\theta} = [\theta_1, ..., \theta_P]^T$ ➔ is the vector $\hat{\boldsymbol{\theta}}$ that maximizes the pdf $f_x(x; \boldsymbol{\theta})$ of the observed data x

- This vector can be found by solving

$$\frac{\partial}{\partial \boldsymbol{\theta}} \ln f_x(x; \boldsymbol{\theta}) = \mathbf{0}$$

- If multiple solutions exist, the MLE is the solution that maximizes the likelihood function $f_x(x; \boldsymbol{\theta})$

- MLE properties ➔ still valid also in the vector case

Example: DC and noise variance

- $x[n] = \theta_1 + w[n], \quad n = 0, \ldots, N-1, \quad \sigma_w^2 = \theta_2$ with $\theta_2 > 0$
- Likelihood function, LLF, and derivative:

$$f_x(x;\boldsymbol{\theta}) = \frac{1}{(2\pi\theta_2)^{N/2}} \exp\left[-\frac{1}{2\theta_2} \sum_{n=0}^{N-1} (x[n] - \theta_1)^2 \right]$$

$$\ln f_x(x;\boldsymbol{\theta}) = -\frac{N}{2} \ln(2\pi\theta_2) - \frac{1}{2\theta_2} \sum_{n=0}^{N-1} (x[n] - \theta_1)^2$$

$$\frac{\partial}{\partial \theta_1} \ln f_x(x;\boldsymbol{\theta}) = \frac{1}{\theta_2} \sum_{n=0}^{N-1} (x[n] - \theta_1)$$

$$\frac{\partial}{\partial \theta_2} \ln f_x(x;\boldsymbol{\theta}) = -\frac{N}{2\theta_2} - \frac{1}{4\theta_2^2} \left[-2 \sum_{n=0}^{N-1} (x[n] - \theta_1)^2 \right]$$

Example: DC and noise variance

- $\dfrac{\partial}{\partial \theta_1} \ln f_x(x;\boldsymbol{\theta}) = 0 \quad \rightarrow \quad \hat{\theta}_1 = \dfrac{1}{N}\sum_{n=0}^{N-1} x[n]$
- $\dfrac{\partial}{\partial \theta_2} \ln f_x(x;\boldsymbol{\theta}) = 0 \quad \rightarrow \quad \hat{\theta}_2 = \dfrac{1}{N}\sum_{n=0}^{N-1} (x[n]-\hat{\theta}_1)^2$
- Note that $E\{\hat{\boldsymbol{\theta}}\} = \begin{bmatrix} \theta_1 \\ \dfrac{N-1}{N}\theta_2 \end{bmatrix} \xrightarrow[N\to\infty]{} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \boldsymbol{\theta}$

$$\mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}} = \begin{bmatrix} \dfrac{\theta_2}{N} & 0 \\ 0 & \dfrac{2(N-1)\theta_2^2}{N^2} \end{bmatrix} \xrightarrow[N\to\infty]{} \begin{bmatrix} \dfrac{\theta_2}{N} & 0 \\ 0 & \dfrac{2\theta_2^2}{N} \end{bmatrix} = \mathbf{I}^{-1}(\boldsymbol{\theta})$$

MLE – linear Gaussian model

- $\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}, \qquad \mathbf{C}_{\mathbf{xx}} = \mathbf{C}_{\mathbf{ww}}, \qquad \mathbf{w}$ Gaussian noise
- MLE ➔ $f_x(\mathbf{x};\boldsymbol{\theta}) = \dfrac{1}{(2\pi)^{N/2}\sqrt{\det(\mathbf{C}_{\mathbf{xx}})}} \exp\left[-\dfrac{1}{2}(\mathbf{x}-\mathbf{H}\boldsymbol{\theta})^T \mathbf{C}_{\mathbf{xx}}^{-1}(\mathbf{x}-\mathbf{H}\boldsymbol{\theta})\right]$

$$\ln f_x(\mathbf{x};\boldsymbol{\theta}) = -\frac{1}{2}\ln\left[(2\pi)^N \det(\mathbf{C}_{\mathbf{xx}})\right] - \frac{1}{2}(\mathbf{x}-\mathbf{H}\boldsymbol{\theta})^T \mathbf{C}_{\mathbf{xx}}^{-1}(\mathbf{x}-\mathbf{H}\boldsymbol{\theta})$$

$$\frac{\partial}{\partial \boldsymbol{\theta}} \ln f_x(\mathbf{x};\boldsymbol{\theta}) = \mathbf{H}^T \mathbf{C}_{\mathbf{xx}}^{-1}(\mathbf{x}-\mathbf{H}\boldsymbol{\theta})$$

- The MLE solution is $\hat{\boldsymbol{\theta}} = (\mathbf{H}^T \mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{H})^{-1}\mathbf{H}^T \mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{x}$

$$\mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}} = (\mathbf{H}^T \mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{H})^{-1}$$

- This coincides with MVU estimator and BLUE



1.5

Least Squares Estimator (LSE)



Least Squares Estimator

- If the target signal is known, but there is no probabilistic model for the measurement error $e[n]$ ➔ a least squares approach may be opportune

- The data model is purely deterministic:

$$x[n] = s[n,\theta] + e[n], \qquad n = 0,\ldots,N-1$$

- Least Squares Estimator (LSE) aims at minimizing the cost function

$$J(\theta) = \sum_{n=0}^{N-1} e^2[n] = \sum_{n=0}^{N-1} \left(x[n] - s[n,\theta] \right)^2$$



Example: LSE for DC level

- Target is constant: $s[n,\theta]=\theta$
- Data model: $x[n]=\theta+e[n], \qquad n=0,\ldots,N-1$
- Cost function: $J(\theta)=\sum_{n=0}^{N-1}\left(x[n]-\theta\right)^2$
- Minimization:

$$\frac{\partial}{\partial\theta}J(\theta)=0 \;\Rightarrow\; -2\sum_{n=0}^{N-1}\left(x[n]-\theta\right)=0 \;\Rightarrow\; \sum_{n=0}^{N-1}x[n]=N\theta$$

- Solution: $\hat{\theta}_{\mathrm{LSE}}=\frac{1}{N}\sum_{n=0}^{N-1}x[n]$



Example: linear LSE

- Target is linear: $s[n,\theta] = \theta h[n]$
- Data model: $x[n] = \theta h[n] + e[n], \qquad n = 0,\ldots,N-1$
- Cost function: $J(\theta) = \sum_{n=0}^{N-1} \big(x[n] - \theta h[n]\big)^2$
- Minimization: $\dfrac{\partial}{\partial\theta} J(\theta) = 0 \;\Rightarrow -2\sum_{n=0}^{N-1} \big(x[n] - \theta h[n]\big) h[n] = 0$
- Solution: $\hat{\theta}_{\mathrm{LSE}} = \dfrac{\sum_{n=0}^{N-1} x[n]h[n]}{\sum_{n=0}^{N-1} h^2[n]}$

Example: linear LSE

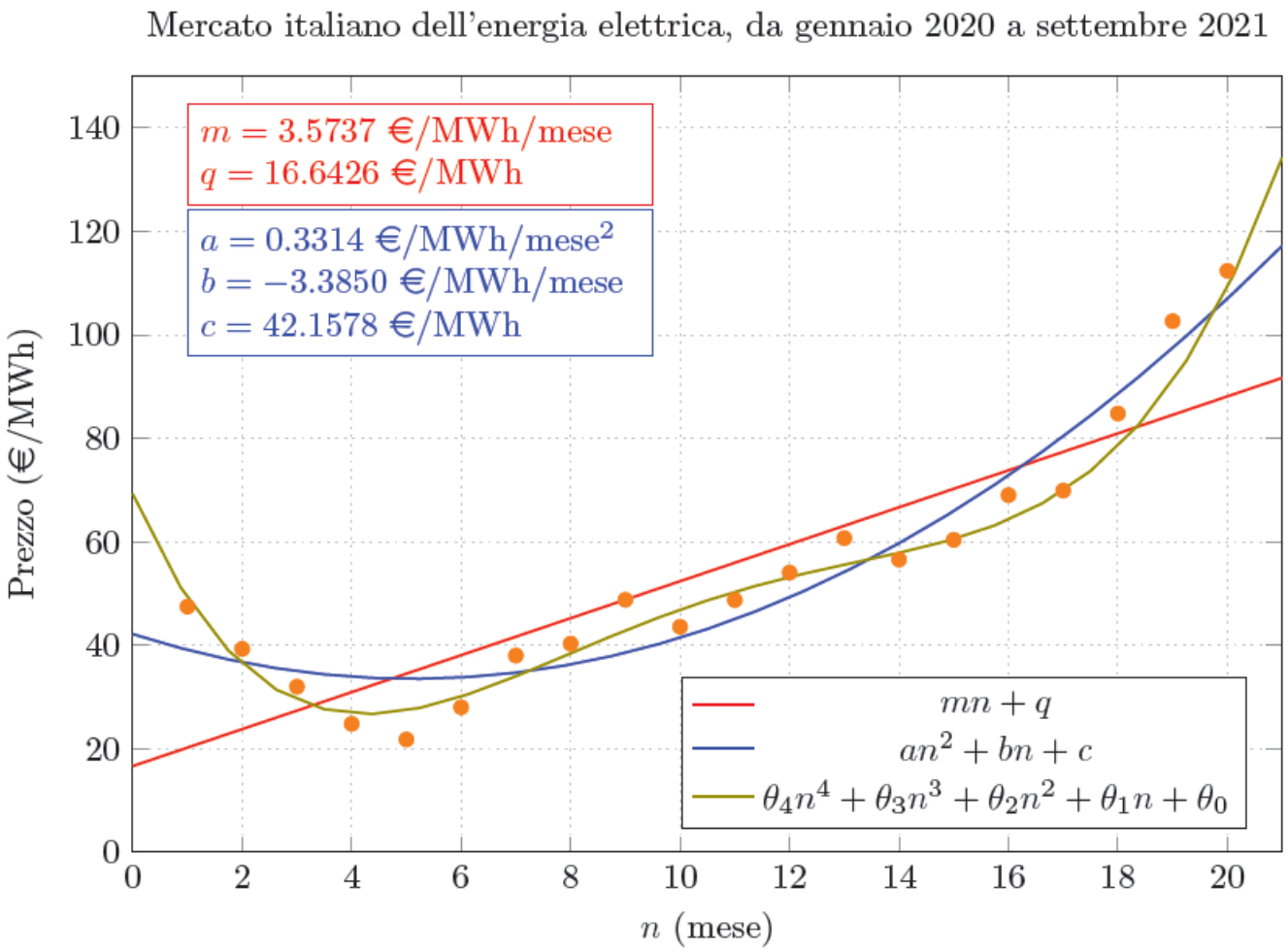


Figura 5.12: Esempio di stima lineare ai minimi quadrati. Fonte dati: GME.



Linear LSE – vector case

- Vector of parameters: $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_P]^T$
- Target vector is linear: $\mathbf{s} = \mathbf{H}\boldsymbol{\theta}$
- Data model: $\mathbf{x} = \mathbf{s} + \mathbf{e} = \mathbf{H}\boldsymbol{\theta} + \mathbf{e}$
- Cost function: $J(\boldsymbol{\theta}) = (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T (\mathbf{x} - \mathbf{H}\boldsymbol{\theta}) =$

$$= \mathbf{x}^T\mathbf{x} - \mathbf{x}^T\mathbf{H}\boldsymbol{\theta} - \boldsymbol{\theta}^T\mathbf{H}^T\mathbf{x} + \boldsymbol{\theta}^T\mathbf{H}^T\mathbf{H}\boldsymbol{\theta}$$

- $\dfrac{\partial}{\partial \boldsymbol{\theta}} J(\boldsymbol{\theta}) = \mathbf{0} \Rightarrow -2\mathbf{H}^T\mathbf{x} + 2\mathbf{H}^T\mathbf{H}\boldsymbol{\theta} = \mathbf{0} \Rightarrow \hat{\boldsymbol{\theta}}_{\mathrm{LSE}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}$



Orthogonality principle

- Linear LSE solution satisfies $\mathbf{H}^T\mathbf{x} - \mathbf{H}^T\mathbf{H}\hat{\boldsymbol{\theta}}_{\text{LSE}} = \mathbf{0}$
- Error vector: $\hat{\mathbf{e}} = \mathbf{x} - \hat{\mathbf{s}} = \mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}_{\text{LSE}}$
- Therefore: $\mathbf{H}^T\mathbf{x} - \mathbf{H}^T\mathbf{H}\hat{\boldsymbol{\theta}}_{\text{LSE}} = \mathbf{H}^T(\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}_{\text{LSE}}) = \mathbf{H}^T\hat{\mathbf{e}} = \mathbf{0}$

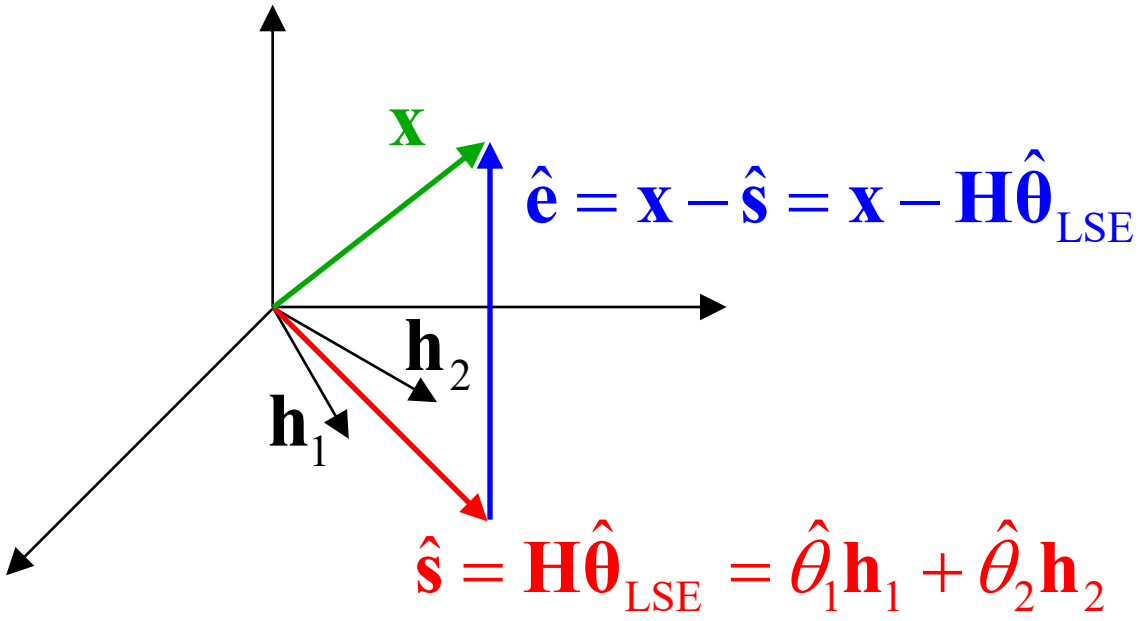
  error vector is orthogonal to the columns of $\mathbf{H}$

- Target vector: $\hat{\mathbf{s}} = \mathbf{H}\hat{\boldsymbol{\theta}}_{\text{LSE}}$

  target vector in the subspace of the columns of $\mathbf{H}$

Geometrical interpretation

- Linear LSE for $P = 2$ when $N = 3$ ➔ $\mathbf{H} = [\mathbf{h}_1 \quad \mathbf{h}_2]$



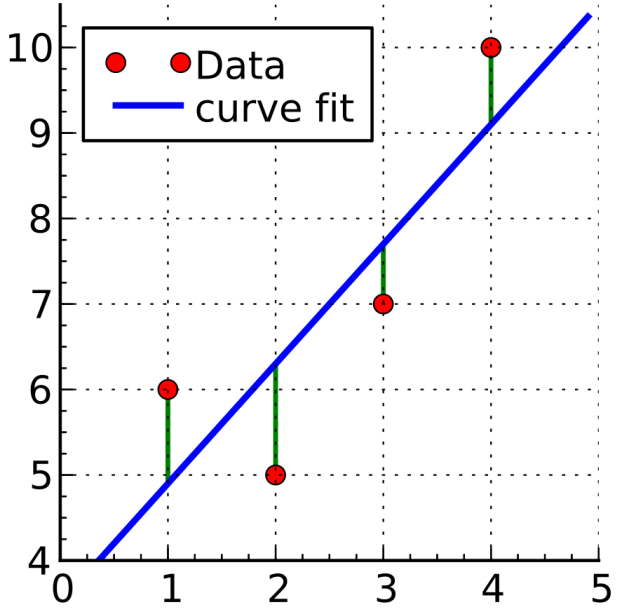
- Target vector $\hat{\mathbf{s}}$ ➔ is the projection of $\mathbf{x}$ onto $\mathbf{H}$

Line fitting

- $x[n] = \theta_1 + \theta_2 n + e[n]$

  $n = 0, \ldots, N-1$

- $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$ $\qquad \mathbf{H} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & N-1 \end{bmatrix}$



- $\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{e}$ ➔ error with no model ➔ line fitting using a linear LSE approach ➔ $\hat{\boldsymbol{\theta}}_{\text{LSE}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}$

- $\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$ ➔ noise with statistical model ➔ line fitting using a CRLB (or a BLUE) approach



PART 2

Bayesian Estimation



Bayesian estimation

- Minimum Mean Square Error (MMSE) Estimation
- Linear MMSE Estimation



Bayesian estimation

- The true unknown parameter θ is assumed a **random variable** whose pdf can be exploited for improving the estimation performance

- The joint pdf of data and unknown parameter is $f_{x\theta}(x[0],...,x[N-1],\theta) = f_{x|\theta}(x[0],...,x[N-1]|\theta) f_{\theta}(\theta)$ where $f_{\theta}(\theta)$ is our **prior knowledge** about the unknown parameter θ to be estimated

- The estimated parameter $\hat{\theta}$ is random and the performance of the estimator is given by the statistics (pdf, mean, variance, MSE) of $\hat{\theta}$



Bayesian MSE

- We want to minimize the Bayesian MSE (BMSE), which is the expected value of the classical MSE

$$\mathrm{BMSE}\{\hat{\theta}\} = E_{\theta}\{\,\mathrm{MSE}\{\hat{\theta}\}\,\}$$

- The BMSE can be interpreted as the statistical average of the classical MSE, where the average is done over the prior knowledge on the unknown parameter θ

- While the classical MSE usually depends on θ, the BMSE does not depend on θ



Reminder

- Definition of conditional pdf:

$$f_{\theta|x}(\theta \mid x) = \frac{f_{x\theta}(x,\theta)}{f_x(x)} \qquad f_{x|\theta}(x \mid \theta) = \frac{f_{x\theta}(x,\theta)}{f_\theta(\theta)}$$

- Bayes Theorem:

$$f_{\theta|x}(\theta \mid x) = \frac{f_{x|\theta}(x \mid \theta) f_\theta(\theta)}{f_x(x)}$$



Bayesian vs classical MSE

- Classical: $\mathrm{MSE}\{\hat{\theta}\} = E_x\{(\hat{\theta}-\theta)^2\} =$

$$= \int_x (\hat{\theta}-\theta)^2 f_x(x;\theta)dx = \mathrm{Var}\{\hat{\theta}\} + b^2(\theta)$$

- Bayesian: $\mathrm{BMSE}\{\hat{\theta}\} = E_{x\theta}\{(\theta-\hat{\theta})^2\} =$

$$= \iint (\theta-\hat{\theta})^2 f_{x\theta}(x,\theta)dxd\theta =$$

$$= \int_\theta \int_x (\theta-\hat{\theta})^2 f_{x|\theta}(x\,|\,\theta)dx f_\theta(\theta)d\theta =$$

$$= E_\theta\{\mathrm{Var}\{\hat{\theta}\}\} + E_\theta\{b^2(\theta)\}$$



2.1

Minimum Mean Square Error (MMSE) Estimation



Minimum Mean Square Error

- Minimum mean square error (MMSE) estimator ➔ obtained by minimizing the BMSE

$$\begin{aligned}\mathrm{BMSE}\{\hat{\theta}\} &= E_{x\theta}\{(\theta-\hat{\theta})^2\} = \\ &= \iint (\theta-\hat{\theta})^2 f_{x\theta}(x,\theta)dxd\theta = \\ &= \int_{\theta}\int_{x} (\theta-\hat{\theta})^2 f_{x|\theta}(x\,|\,\theta)dx f_{\theta}(\theta)d\theta = \\ &= \int_{x}\int_{\theta} (\theta-\hat{\theta})^2 f_{\theta|x}(\theta\,|\,x)d\theta f_{x}(x)dx\end{aligned}$$

- $f_x(x) \geq 0$ ➔ the BMSE is minimized when the inner integral in red is minimized

MMSE estimator

$$\frac{\partial}{\partial\hat{\theta}}\int_{\theta}(\theta-\hat{\theta})^2 f_{\theta|x}(\theta\,|\,x)d\theta=0$$

$$\int_{\theta}\frac{\partial}{\partial\hat{\theta}}(\theta-\hat{\theta})^2 f_{\theta|x}(\theta\,|\,x)d\theta=0$$

$$-2\int_{\theta}(\theta-\hat{\theta}) f_{\theta|x}(\theta\,|\,x)d\theta=0$$

$$\int_{\theta}\hat{\theta} f_{\theta|x}(\theta\,|\,x)d\theta=\int_{\theta}\theta f_{\theta|x}(\theta\,|\,x)d\theta$$

$$\hat{\theta}\int_{\theta} f_{\theta|x}(\theta\,|\,x)d\theta=\int_{\theta}\theta f_{\theta|x}(\theta\,|\,x)d\theta$$

$$\hat{\theta}_{\mathrm{MMSE}}=\int_{\theta}\theta f_{\theta|x}(\theta\,|\,x)d\theta$$



MMSE estimator

- Hence the MMSE estimator is a conditional mean estimator:

$$\hat{\theta}_{\text{MMSE}} = E_{\theta|x}\{\theta \mid x\} = \mu_{\theta|x} =$$

$$= \int_{\theta} \theta f_{\theta|x}(\theta \mid x) d\theta =$$

$$= \int_{\theta} \theta \frac{f_{x\theta}(x,\theta)}{f_x(x)} d\theta = \frac{\int_{\theta} \theta f_{x\theta}(x,\theta) d\theta}{f_x(x)}$$

- The MMSE estimator always exists



Reminder

- Prior pdf (**before** the observation of data): $f_\theta(\theta)$

- Posterior pdf (**after** the observation of data):

$$f_{\theta|x}(\theta \mid x) = \frac{f_{x\theta}(x,\theta)}{f_x(x)} = \frac{f_{x|\theta}(x \mid \theta) f_\theta(\theta)}{f_x(x)} =$$

$$= \frac{f_{x|\theta}(x \mid \theta) f_\theta(\theta)}{\int f_{x|\theta}(x \mid \theta) f_\theta(\theta) d\theta}$$

- Joint pdf: $f_{x\theta}(x,\theta) \neq f_x(x) f_\theta(\theta)$



Finding the MMSE estimator

- 1st step: find the joint pdf $f_{x\theta}(x,\theta)$
- 2st step: saturation $f_x(x) = \int_{\theta} f_{x\theta}(x,\theta)d\theta$
- 3rd step: compute the posterior pdf

$$f_{\theta|x}(\theta \mid x) = \frac{f_{x\theta}(x,\theta)}{f_x(x)}$$

- 4th step: compute the conditional mean

$$\hat{\theta}_{\text{MMSE}} = \int_{\theta} \theta f_{\theta|x}(\theta \mid x)d\theta$$

Example: DC level with Gaussian prior pdf

- $x[n] = \theta + w[n], \quad n = 0, \ldots, N-1, \qquad w[n]$ AWGN,

  θ Gaussian with mean μ_θ and variance σ_θ^2

- The posterior pdf is Gaussian:

$$f_x(\theta \mid x) = \frac{1}{(2\pi\sigma_{\theta|x}^2)^{1/2}} \exp\left[-\frac{(\theta - \mu_{\theta|x})^2}{2\sigma_{\theta|x}^2}\right]$$

with

$$\mu_{\theta|x} = \frac{\frac{\sigma_w^2}{N}\mu_\theta + \sigma_\theta^2 \frac{1}{N}\sum_{n=0}^{N-1} x[n]}{\sigma_\theta^2 + \frac{\sigma_w^2}{N}} \qquad \sigma_{\theta|x}^2 = \frac{\sigma_\theta^2 \frac{\sigma_w^2}{N}}{\sigma_\theta^2 + \frac{\sigma_w^2}{N}}$$

Example: DC level with Gaussian prior pdf

- The MMSE estimator can be expressed as

$$\hat{\theta}_{\text{MMSE}} = \mu_{\theta|x} = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \frac{\sigma_w^2}{N}} \left[\frac{1}{N} \sum_{n=0}^{N-1} x[n] \right] + \frac{\frac{\sigma_w^2}{N}}{\sigma_\theta^2 + \frac{\sigma_w^2}{N}} \mu_\theta$$

- The MMSE estimator is a weighted average of the sample mean and the prior mean μ_θ

- When the number of samples increases, the MMSE estimator tends to the sample mean



Example: DC level with Gaussian prior pdf

- The BMSE of the MMSE estimator is

$$\mathrm{BMSE}\{\hat{\theta}_{\mathrm{MMSE}}\} = \int_x \mathrm{Var}\{\hat{\theta}_{\mathrm{MMSE}}\} f_x(x)dx = \int_x \sigma^2_{\theta|x} f_x(x)dx = \sigma^2_{\theta|x}$$

- BMSE: lower than MSE of sample mean estimator

$$\sigma^2_{\theta|x} = \frac{\sigma^2_\theta}{\sigma^2_\theta + \frac{\sigma^2_w}{N}} \cdot \frac{\sigma^2_w}{N} \leq \frac{\sigma^2_w}{N} = \mathrm{MSE}\{\hat{\theta}_{\mathrm{MVU}}\}$$

and lower than prior variance $\sigma^2_{\theta|x} = \dfrac{\frac{\sigma^2_w}{N}}{\sigma^2_\theta + \frac{\sigma^2_w}{N}} \cdot \sigma^2_\theta \leq \sigma^2_\theta$



Proof of $\mathrm{BMSE}\{\hat{\theta}_{\mathrm{MMSE}}\} = \int \sigma^2_{\theta|x} f_x(x)dx$

$$\mathrm{BMSE}\{\hat{\theta}_{\mathrm{MMSE}}\} = E_{x\theta}\{(\theta - \hat{\theta}_{\mathrm{MMSE}})^2\} =$$

$$= \int_x \int_\theta (\theta - \hat{\theta}_{\mathrm{MMSE}})^2 f_{x\theta}(x,\theta)d\theta dx =$$

$$= \int_x \int_\theta (\theta - \hat{\theta}_{\mathrm{MMSE}})^2 f_{\theta|x}(\theta \mid x)d\theta f_x(x)dx =$$

$$= \int_x \mathrm{Var}\{\hat{\theta}_{\mathrm{MMSE}}\} f_x(x)dx =$$

$$= \int_x \sigma^2_{\theta|x} f_x(x)dx$$



MMSE estimator – vector case

- Vector of parameters: $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_P]^T$
- MMSE estimators (scalar): $\hat{\theta}_{i,\mathrm{MMSE}} = \int_{\theta_i} \theta_i f_{\theta_i|x}(\theta_i \mid x) d\theta_i$

  where $f_{\theta_i|x}(\theta_i \mid x) = \int \ldots \int f_{\boldsymbol{\theta}|\mathbf{x}}(\boldsymbol{\theta} \mid \mathbf{x}) d\theta_1 \ldots d\theta_{i-1} d\theta_{i+1} \ldots d\theta_P$
- By substitution: $\hat{\theta}_{i,\mathrm{MMSE}} = \int_{\boldsymbol{\theta}} \theta_i f_{\boldsymbol{\theta}|\mathbf{x}}(\boldsymbol{\theta} \mid \mathbf{x}) d\boldsymbol{\theta}$
- Therefore: $\hat{\boldsymbol{\theta}}_{\mathrm{MMSE}} = \int_{\boldsymbol{\theta}} \boldsymbol{\theta} f_{\boldsymbol{\theta}|\mathbf{x}}(\boldsymbol{\theta} \mid \mathbf{x}) d\boldsymbol{\theta} = E\{\boldsymbol{\theta} \mid \mathbf{x}\}$
- $\mathrm{BMSE}\{\hat{\theta}_{i,\mathrm{MMSE}}\} = \int_{\mathbf{x}} [\mathbf{C}_{\boldsymbol{\theta}|\mathbf{x}}]_{i,i} f_{\mathbf{x}}(\mathbf{x}) d\mathbf{x}$ $\qquad \mathbf{C}_{\boldsymbol{\theta}|\mathbf{x}} = \mathrm{Cov}\{\boldsymbol{\theta} \mid \mathbf{x}\}$

Linear Gaussian model

- This is the vector case of the previous example:

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

- The vectors $\boldsymbol{\theta}$ and $\mathbf{w}$ are jointly Gaussian, are independent, and have mean and covariance given by $(\boldsymbol{\mu}_{\theta}, \mathbf{C}_{\theta\theta})$ and $(\mathbf{0}, \mathbf{C}_{\mathbf{ww}})$ respectively

- The posterior pdf is also Gaussian, with mean and covariance given by

$$\boldsymbol{\mu}_{\boldsymbol{\theta}|\mathbf{x}} = E\{\boldsymbol{\theta} \mid \mathbf{x}\} = \boldsymbol{\mu}_{\boldsymbol{\theta}} + \mathbf{C}_{\theta\theta}\mathbf{H}^T(\mathbf{H}\mathbf{C}_{\theta\theta}\mathbf{H}^T + \mathbf{C}_{\mathbf{ww}})^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\mu}_{\boldsymbol{\theta}})$$

$$\mathbf{C}_{\boldsymbol{\theta}|\mathbf{x}} = \mathrm{Cov}\{\boldsymbol{\theta} \mid \mathbf{x}\} = \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta\theta}\mathbf{H}^T(\mathbf{H}\mathbf{C}_{\theta\theta}\mathbf{H}^T + \mathbf{C}_{\mathbf{ww}})^{-1}\mathbf{H}\mathbf{C}_{\theta\theta}$$



Linear Gaussian model

- MMSE estimator ➔ is exactly the mean of the posterior pdf

$$\hat{\boldsymbol{\theta}}_{\text{MMSE}} = E\{\boldsymbol{\theta} \mid \mathbf{x}\} = \boldsymbol{\mu}_{\boldsymbol{\theta}|\mathbf{x}} =$$

$$= \boldsymbol{\mu}_{\boldsymbol{\theta}} + \mathbf{C}_{\boldsymbol{\theta}\boldsymbol{\theta}}\mathbf{H}^T(\mathbf{H}\mathbf{C}_{\boldsymbol{\theta}\boldsymbol{\theta}}\mathbf{H}^T + \mathbf{C}_{\mathbf{ww}})^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\mu}_{\boldsymbol{\theta}})$$

- BMSE ➔ is obtained by averaging the variance of the posterior pdf over the pdf of the data ➔ the BMSE for the ith MMSE estimator is given by the element (i,i) of the covariance of the posterior pdf

$$\text{BMSE}\{\hat{\theta}_{i,\text{MMSE}}\} = \int_{\mathbf{x}} [\mathbf{C}_{\boldsymbol{\theta}|\mathbf{x}}]_{i,i} f_{\mathbf{x}}(\mathbf{x}) d\mathbf{x} = [\mathbf{C}_{\boldsymbol{\theta}|\mathbf{x}}]_{i,i}$$



Is the MMSE estimator biased?

- If we define the bias in a Bayesian sense ➔ the MMSE estimator is unbiased:

$$E_{x\theta}\{\theta-\hat{\theta}_{\text{MMSE}}\}=E_{x\theta}\{\theta\}-E_{x\theta}\left\{\int_{\theta}\theta f_{\theta|x}(\theta\,|\,x)d\theta\right\}=$$

$$=E_{x}\{E_{\theta|x}\{\theta\}\}-E_{x}\left\{E_{\theta|x}\left\{\int_{\theta}\theta f_{\theta|x}(\theta\,|\,x)d\theta\right\}\right\}=$$

$$=E_{x}\left\{\int_{\theta}\theta f_{\theta|x}(\theta\,|\,x)d\theta\right\}-E_{x}\left\{\int_{\theta}\theta f_{\theta|x}(\theta\,|\,x)d\theta\right\}=0$$

- If we define the bias in a classical sense ➔ the MMSE estimator is (conditionally) biased:

$$E_{x|\theta}\{\hat{\theta}_{\text{MMSE}}-\theta\}=E_{x|\theta}\{\hat{\theta}_{\text{MMSE}}\}-E_{x|\theta}\{\theta\}=K-\theta\neq 0$$



2.2

Linear MMSE (LMMSE) Estimation

Linear MMSE estimator

- Linear MMSE (LMMSE) estimator ➔ minimizes the BMSE subject to the linear (affine) constraint

$$\hat{\theta} = a_N + \sum_{n=0}^{N-1} a_n x[n] = a_N + \mathbf{a}^T \mathbf{x}$$

where $\mathbf{x} = \left[x[0], x[1], ..., x[N-1]\right]^T$, $\mathbf{a} = [a_0, a_1, ..., a_{N-1}]^T$

- Minimization of $\mathrm{BMSE}\{\hat{\theta}\} = E_{x\theta}\{(\theta - \hat{\theta})^2\}$

$$= E_{x\theta}\left\{\left(\theta - a_N - \sum_{n=0}^{N-1} a_n x[n]\right)^2\right\}$$

$$= E_{x\theta}\{(\theta - a_N - \mathbf{a}^T \mathbf{x})^2\}$$



LMMSE estimator

- Last parameter ➔ $\frac{\partial}{\partial a_N}\mathrm{BMSE}\{\hat{\theta}\}=0$

$$\frac{\partial}{\partial a_N}E_{x\theta}\{(\theta-a_N-\mathbf{a}^T\mathbf{x})^2\}=0$$

$$E_{x\theta}\left\{\frac{\partial}{\partial a_N}(\theta-a_N-\mathbf{a}^T\mathbf{x})^2\right\}=0$$

$$E_{x\theta}\{-2(\theta-a_N-\mathbf{a}^T\mathbf{x})\}=0$$

$$E_{\theta}\{\theta\}-a_N-\mathbf{a}^T E_x\{\mathbf{x}\}=0$$

- Therefore $a_N=E_{\theta}\{\theta\}-\mathbf{a}^T E_{\mathbf{x}}\{\mathbf{x}\}$ ➔ $\boxed{a_N=\mu_{\theta}-\mathbf{a}^T\boldsymbol{\mu}_{\mathbf{x}}}$

LMMSE estimator

- To find the other parameters ➔ substitution ➔

$$\mathrm{BMSE}\{\hat{\theta}\} = E_{x\theta}\{[\theta - \mu_\theta - \mathbf{a}^T(\mathbf{x} - \boldsymbol{\mu}_\mathbf{x})]^2\} =$$

$$= E_{x\theta}\{(\theta - \mu_\theta)^2\} - 2\mathbf{a}^T E_{x\theta}\{(\mathbf{x} - \boldsymbol{\mu}_\mathbf{x})(\theta - \mu_\theta)\} +$$

$$+ \mathbf{a}^T E_{\mathbf{xx}}\{(\mathbf{x} - \boldsymbol{\mu}_\mathbf{x})(\mathbf{x} - \boldsymbol{\mu}_\mathbf{x})^T\}\mathbf{a} =$$

$$= \sigma_\theta^2 - 2\mathbf{a}^T \mathbf{c}_{\mathbf{x}\theta} + \mathbf{a}^T \mathbf{C}_{\mathbf{xx}}\mathbf{a}$$

- Optimal **a** ➔ $\dfrac{\partial}{\partial \mathbf{a}}\mathrm{BMSE}\{\hat{\theta}\} = \dfrac{\partial}{\partial \mathbf{a}}[\sigma_\theta^2 - 2\mathbf{a}^T \mathbf{c}_{\mathbf{x}\theta} + \mathbf{a}^T \mathbf{C}_{\mathbf{xx}}\mathbf{a}] = \mathbf{0}$

- Hence ➔ $-2\mathbf{c}_{\mathbf{x}\theta} + 2\mathbf{C}_{\mathbf{xx}}\mathbf{a} = \mathbf{0}$ ➔ $\boxed{\mathbf{a} = \mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{c}_{\mathbf{x}\theta}}$



LMMSE estimator

- Final solution for the LMMSE estimator ➔

$$\begin{aligned}\hat{\theta}_{\text{LMMSE}} &= a_N + \mathbf{a}^T\mathbf{x} = \\ &= \mu_\theta - \mathbf{a}^T\boldsymbol{\mu}_\mathbf{x} + \mathbf{a}^T\mathbf{x} = \\ &= \mu_\theta + \mathbf{a}^T(\mathbf{x} - \boldsymbol{\mu}_\mathbf{x}) = \\ &= \mu_\theta + \mathbf{c}_{\mathbf{x}\theta}^T\mathbf{C}_{\mathbf{xx}}^{-1}(\mathbf{x} - \boldsymbol{\mu}_\mathbf{x})\end{aligned}$$

- The LMMSE estimator **depends only on the first two moments** of the joint pdf:

$$E\left\{\begin{bmatrix}\mathbf{x}\\ \theta\end{bmatrix}\right\} = \begin{bmatrix}\boldsymbol{\mu}_\mathbf{x}\\ \mu_\theta\end{bmatrix} \qquad \text{Cov}\left\{\begin{bmatrix}\mathbf{x}\\ \theta\end{bmatrix}\right\} = \begin{bmatrix}\mathbf{C}_{\mathbf{xx}} & \mathbf{c}_{\mathbf{x}\theta}\\ \mathbf{c}_{\mathbf{x}\theta}^T & \sigma_\theta^2\end{bmatrix}$$



LMMSE estimator

- The BMSE of the LMMSE estimator is

$$\begin{aligned}\mathrm{BMSE}\{\hat{\theta}_{\mathrm{LMMSE}}\} &= \sigma_\theta^2 - 2\mathbf{a}^T\mathbf{c}_{\mathbf{x}\theta} + \mathbf{a}^T\mathbf{C}_{\mathbf{xx}}\mathbf{a} = \\ &= \sigma_\theta^2 - 2(\mathbf{c}_{\mathbf{x}\theta}^T\mathbf{C}_{\mathbf{xx}}^{-1})\mathbf{c}_{\mathbf{x}\theta} + (\mathbf{c}_{\mathbf{x}\theta}^T\mathbf{C}_{\mathbf{xx}}^{-1})\mathbf{C}_{\mathbf{xx}}(\mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{c}_{\mathbf{x}\theta}) = \\ &= \sigma_\theta^2 - \mathbf{c}_{\mathbf{x}\theta}^T\mathbf{C}_{\mathbf{xx}}^{-1}\mathbf{c}_{\mathbf{x}\theta}\end{aligned}$$

- Note that $\mathrm{BMSE}\{\hat{\theta}_{\mathrm{LMMSE}}\} \leq \sigma_\theta^2$
- Joint pdf is Gaussian ➔ LMMSE estimator and MMSE estimator coincide ➔ same BMSE
- Joint pdf is not Gaussian ➔ LMMSE estimator is suboptimal (larger BMSE), but easier to compute



Example: DC level with uniform prior pdf

- $x[n] = \theta + w[n], \quad n = 0, \ldots, N-1, \qquad w[n]$ AWGN,

  θ uniform between $-A_0$ and A_0

- Mean value $\mu_\theta = 0$, variance $\sigma_\theta^2 = E\{\theta^2\} = \dfrac{A_0^2}{3}$
- Vector model: $\mathbf{x} = \theta \mathbf{1}_N + \mathbf{w}$
- LMMSE estimator: $\hat{\theta}_{\text{LMMSE}} = \mu_\theta + \mathbf{c}_{\mathbf{x}\theta}^T \mathbf{C}_{\mathbf{xx}}^{-1} (\mathbf{x} - \boldsymbol{\mu}_\mathbf{x})$
- We must evaluate $\boldsymbol{\mu}_\mathbf{x}, \ \mathbf{c}_{\mathbf{x}\theta}, \ \mathbf{C}_{\mathbf{xx}}^{-1}$

Example: DC level with uniform prior pdf

- $\mathbf{x} = \theta \mathbf{1}_N + \mathbf{w}$, $\hat{\theta}_{\text{LMMSE}} = \mathbf{c}_{\mathbf{x}\theta}^T \mathbf{C}_{\mathbf{xx}}^{-1} (\mathbf{x} - \boldsymbol{\mu}_{\mathbf{x}})$

- $\boldsymbol{\mu}_{\mathbf{x}} = E\{\mathbf{x}\} = E\{\theta\}\mathbf{1}_N + E\{\mathbf{w}\} = \mathbf{0}_N$

$$\mathbf{c}_{\mathbf{x}\theta} = E\{\mathbf{x}\theta\} = E\{(\theta \mathbf{1}_N + \mathbf{w})\theta\} = E\{\theta^2\}\mathbf{1}_N + E\{\mathbf{w}\}E\{\theta\} = \sigma_\theta^2 \mathbf{1}_N$$

$$\mathbf{C}_{\mathbf{xx}} = E\{\mathbf{x}\mathbf{x}^T\} = E\{(\theta \mathbf{1}_N + \mathbf{w})(\theta \mathbf{1}_N + \mathbf{w})^T\} =$$
$$= E\{\theta^2\}\mathbf{1}_N \mathbf{1}_N^T + E\{\mathbf{w}\mathbf{w}^T\} = \sigma_\theta^2 \mathbf{1}_N \mathbf{1}_N^T + \sigma_w^2 \mathbf{I}_N$$

- To find $\mathbf{C}_{\mathbf{xx}}^{-1}$, let us write $\mathbf{C}_{\mathbf{xx}} = \sigma_w^2 \left(\mathbf{I}_N + \frac{\sigma_\theta^2}{\sigma_w^2} \mathbf{1}_N \mathbf{1}_N^T \right)$

Example: DC level with uniform prior pdf

- $\mathbf{x} = \theta \mathbf{1}_N + \mathbf{w}$, $\hat{\theta}_{\text{LMMSE}} = \mathbf{c}_{\mathbf{x}\theta}^T \mathbf{C}_{\mathbf{xx}}^{-1} \mathbf{x} = \sigma_\theta^2 \mathbf{1}_N^T \mathbf{C}_{\mathbf{xx}}^{-1} \mathbf{x}$
- Using $\left(\mathbf{I}_N + \mathbf{u}\mathbf{v}^T\right)^{-1} = \mathbf{I}_N - \frac{1}{1+\mathbf{v}^T\mathbf{u}}\mathbf{u}\mathbf{v}^T$, we obtain

$$\mathbf{C}_{\mathbf{xx}}^{-1} = \sigma_w^{-2}\left(\mathbf{I}_N + \frac{\sigma_\theta^2}{\sigma_w^2}\mathbf{1}_N\mathbf{1}_N^T\right)^{-1} = \sigma_w^{-2}\left[\mathbf{I}_N - \frac{1}{1+\mathbf{1}_N^T\left(\frac{\sigma_\theta^2}{\sigma_w^2}\mathbf{1}_N\right)}\left(\frac{\sigma_\theta^2}{\sigma_w^2}\mathbf{1}_N\right)\mathbf{1}_N^T\right] =$$

$$= \sigma_w^{-2}\left(\mathbf{I}_N - \frac{\sigma_\theta^2}{\sigma_w^2 + \sigma_\theta^2\mathbf{1}_N^T\mathbf{1}_N}\mathbf{1}_N\mathbf{1}_N^T\right) = \sigma_w^{-2}\mathbf{I}_N - \frac{\sigma_w^{-2}\sigma_\theta^2}{\sigma_w^2 + \sigma_\theta^2 N}\mathbf{1}_N\mathbf{1}_N^T$$

Example: DC level with uniform prior pdf

- $\mathbf{x} = \theta \mathbf{1}_N + \mathbf{w}$, $\hat{\theta}_{\text{LMMSE}} = \mathbf{c}_{\mathbf{x}\theta}^T \mathbf{C}_{\mathbf{xx}}^{-1} \mathbf{x} = \sigma_\theta^2 \mathbf{1}_N^T \mathbf{C}_{\mathbf{xx}}^{-1} \mathbf{x}$
- Since $\mathbf{C}_{\mathbf{xx}}^{-1} = \sigma_w^{-2} \mathbf{I}_N - \frac{\sigma_w^{-2}\sigma_\theta^2}{\sigma_w^2 + \sigma_\theta^2 N} \mathbf{1}_N \mathbf{1}_N^T$, we obtain

$$\hat{\theta}_{\text{LMMSE}} = \sigma_\theta^2 \mathbf{1}_N^T \mathbf{C}_{\mathbf{xx}}^{-1} \mathbf{x} = \sigma_\theta^2 \mathbf{1}_N^T \left(\sigma_w^{-2} \mathbf{I}_N - \frac{\sigma_w^{-2}\sigma_\theta^2}{\sigma_w^2 + \sigma_\theta^2 N} \mathbf{1}_N \mathbf{1}_N^T \right) \mathbf{x} =$$

$$= \left(\frac{\sigma_\theta^2}{\sigma_w^2} \mathbf{1}_N^T \mathbf{I}_N - \frac{\sigma_w^{-2}\sigma_\theta^4}{\sigma_w^2 + \sigma_\theta^2 N} \mathbf{1}_N^T \mathbf{1}_N \mathbf{1}_N^T \right) \mathbf{x} = \left(\frac{\sigma_\theta^2}{\sigma_w^2} \mathbf{1}_N^T - \frac{\sigma_w^{-2}\sigma_\theta^4}{\sigma_w^2 + \sigma_\theta^2 N} N \mathbf{1}_N^T \right) \mathbf{x} =$$

$$= \left(\frac{\sigma_\theta^2}{\sigma_w^2} - \frac{\sigma_w^{-2}\sigma_\theta^4 N}{\sigma_w^2 + \sigma_\theta^2 N} \right) \mathbf{1}_N^T \mathbf{x} = \frac{\sigma_\theta^2}{\sigma_w^2 + \sigma_\theta^2 N} \mathbf{1}_N^T \mathbf{x} = \frac{\sigma_\theta^2}{\frac{\sigma_w^2}{N} + \sigma_\theta^2} \left(\frac{\mathbf{1}_N^T \mathbf{x}}{N} \right)$$

Example: DC level with uniform prior pdf

- $\mathbf{x} = \theta \mathbf{1}_N + \mathbf{w}$

- In this case, the MMSE estimator is nonlinear

$$\hat{\theta}_{\text{MMSE}} = E_{\theta|x}\{\theta \mid x\} = \int_{\theta} \theta f_{\theta|x}(\theta \mid x) d\theta =$$

$$= \frac{\int_{-A_0}^{A_0} \theta \exp\left[-\frac{1}{2\sigma_w^2}\sum_{n=0}^{N-1}(x[n]-\theta)^2\right] d\theta}{\int_{-A_0}^{A_0} \exp\left[-\frac{1}{2\sigma_w^2}\sum_{n=0}^{N-1}(x[n]-\theta)^2\right] d\theta} = \frac{\int_{-A_0}^{A_0} \theta \exp\left[-\frac{N}{2\sigma_w^2}\left(\theta - \frac{\mathbf{1}_N^T \mathbf{x}}{N}\right)^2\right] d\theta}{\int_{-A_0}^{A_0} \exp\left[-\frac{N}{2\sigma_w^2}\left(\theta - \frac{\mathbf{1}_N^T \mathbf{x}}{N}\right)^2\right] d\theta}$$

- No closed form ➔ LMMSE is easier to compute

LMMSE estimator – vector case

- Vector of parameters: $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_P]^T$
- Scalar LMMSE estimators: $\hat{\theta}_{i,\mathrm{LMMSE}} = \mu_{\theta_i} + \mathbf{c}_{\mathbf{x}\theta_i}^T \mathbf{C}_{\mathbf{xx}}^{-1} (\mathbf{x} - \boldsymbol{\mu}_{\mathbf{x}})$
- We have: $\mathbf{C}_{\mathbf{x}\boldsymbol{\theta}} = \mathrm{Cov}\{\mathbf{x}, \boldsymbol{\theta}\} = [\mathbf{c}_{\mathbf{x}\theta_1}, \ldots, \mathbf{c}_{\mathbf{x}\theta_P}]$
- Therefore: $\hat{\boldsymbol{\theta}}_{\mathrm{LMMSE}} = \boldsymbol{\mu}_{\boldsymbol{\theta}} + \mathbf{C}_{\mathbf{x}\boldsymbol{\theta}}^T \mathbf{C}_{\mathbf{xx}}^{-1} (\mathbf{x} - \boldsymbol{\mu}_{\mathbf{x}})$
- $\mathrm{BMSE}\{\hat{\theta}_{i,\mathrm{LMMSE}}\} = \sigma_{\theta_i}^2 - \mathbf{c}_{\mathbf{x}\theta_i}^T \mathbf{C}_{\mathbf{xx}}^{-1} \mathbf{c}_{\mathbf{x}\theta_i} = [\mathbf{C}_{\boldsymbol{\theta}\boldsymbol{\theta}} - \mathbf{C}_{\mathbf{x}\boldsymbol{\theta}}^T \mathbf{C}_{\mathbf{xx}}^{-1} \mathbf{C}_{\mathbf{x}\boldsymbol{\theta}}]_{i,i}$

  where $\mathbf{C}_{\boldsymbol{\theta}\boldsymbol{\theta}} = \mathrm{Cov}\{\boldsymbol{\theta}\}$



Linear model

- The vector model is linear: $\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$
- The vectors $\boldsymbol{\theta}$ and $\mathbf{w}$ are uncorrelated, with mean and covariance $(\boldsymbol{\mu}_{\boldsymbol{\theta}}, \mathbf{C}_{\boldsymbol{\theta\theta}})$ and $(\mathbf{0}, \mathbf{C}_{\mathbf{ww}})$
- The LMMSE is obtained as $\hat{\boldsymbol{\theta}}_{\text{LMMSE}} = \boldsymbol{\mu}_{\boldsymbol{\theta}} + \mathbf{C}_{\mathbf{x}\boldsymbol{\theta}}^{T} \mathbf{C}_{\mathbf{xx}}^{-1} (\mathbf{x} - \boldsymbol{\mu}_{\mathbf{x}})$
- Use $\boldsymbol{\mu}_{\mathbf{x}} = \mathbf{H}\boldsymbol{\mu}_{\boldsymbol{\theta}}$, $\mathbf{C}_{\mathbf{x}\boldsymbol{\theta}} = \mathbf{H}\mathbf{C}_{\boldsymbol{\theta\theta}}$, $\mathbf{C}_{\mathbf{xx}} = \mathbf{H}\mathbf{C}_{\boldsymbol{\theta\theta}}\mathbf{H}^{T} + \mathbf{C}_{\mathbf{ww}}$
- $$\hat{\boldsymbol{\theta}}_{\text{LMMSE}} = \boldsymbol{\mu}_{\boldsymbol{\theta}} + \mathbf{C}_{\boldsymbol{\theta\theta}}\mathbf{H}^{T} (\mathbf{H}\mathbf{C}_{\boldsymbol{\theta\theta}}\mathbf{H}^{T} + \mathbf{C}_{\mathbf{ww}})^{-1} (\mathbf{x} - \mathbf{H}\boldsymbol{\mu}_{\boldsymbol{\theta}}) =$$
$$= \boldsymbol{\mu}_{\boldsymbol{\theta}} + (\mathbf{C}_{\boldsymbol{\theta\theta}}^{-1} + \mathbf{H}^{T}\mathbf{C}_{\mathbf{ww}}^{-1}\mathbf{H})^{-1} \mathbf{H}^{T}\mathbf{C}_{\mathbf{ww}}^{-1} (\mathbf{x} - \mathbf{H}\boldsymbol{\mu}_{\boldsymbol{\theta}})$$



Linear model

- Two forms of the LMMSE estimator $\hat{\boldsymbol{\theta}}_{\text{LMMSE}}$ ➔ the inverse matrices may have different size

- $$\text{BMSE}\{\boldsymbol{\theta}_{i,\text{LMMSE}}\} = [\mathbf{C}_{\boldsymbol{\theta}\boldsymbol{\theta}} - \mathbf{C}_{\boldsymbol{\theta}\boldsymbol{\theta}}\mathbf{H}^T(\mathbf{H}\mathbf{C}_{\boldsymbol{\theta}\boldsymbol{\theta}}\mathbf{H}^T + \mathbf{C}_{\mathbf{ww}})^{-1}\mathbf{H}\mathbf{C}_{\boldsymbol{\theta}\boldsymbol{\theta}}]_{i,i} =$$
$$= \left[(\mathbf{C}_{\boldsymbol{\theta}\boldsymbol{\theta}}^{-1} + \mathbf{H}^T\mathbf{C}_{\mathbf{ww}}^{-1}\mathbf{H})^{-1}\right]_{i,i}$$

- Jointly Gaussian vectors $\boldsymbol{\theta}$ and $\mathbf{w}$ ➔ LMMSE and MMSE estimators are identical

- Non-Gaussian vectors ➔ LMMSE has larger BMSE, but is easier to compute (no integral, depends on the first two moments only)



Final Summary

Summary of estimators

- If the **prior pdf** of the unknown parameter **is not available** ➔ choose a classical approach

  - First, check if the CRLB can lead to the MVU estimator
  - Alternatively, use a linear estimator ➔ BLUE
  - For large data records ➔ MLE is asymptotically optimal
  - If statistical model of the error is absent ➔ LSE

- When the **prior pdf** of the unknown parameter **is available** ➔ a Bayesian approach is preferable

  - Best accuracy ➔ MMSE estimator
  - Low complexity ➔ LMMSE estimator